LEARNING MODULE

24

Bayesian Inference and Shrinkage

A prior turns a past mean into a weighted average of the data and a belief. Twenty-eight sector means carry almost nothing beyond their average, and fractional Kelly is the Bayes bet of a stated sceptic.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM4 · LM10 · LM11 · LM13 · LM15 · LM16 · LM17 · LM19 · LM20 · LM21

LEADS TO LM26 · LM30 · LM33 · LM38 · LM42 · LM44 · LM45

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 apply Bayes' rule to an unknown parameter, and compute the beta–binomial posterior of a success probability and the data's weight in it
☐	2 derive the posterior of a normal mean with known variance, and explain why the posterior precision is the prior precision plus the data precision
☐	3 distinguish a credible interval from a confidence interval, and compute both for the TAIEX's mean return
☐	4 state the James–Stein estimator with its conditions, and derive its empirical-Bayes reading as an estimated data's weight
☐	5 evaluate a shrinkage estimator out of sample on two halves of a sample, and explain what correlated estimates do to it
☐	6 derive the Bayes Kelly fraction under log utility in a stated model, and compute the prior that makes fractional Kelly the Bayes bet
☐	7 explain what breaks the Bayes Kelly result, and measure the size of its quadratic approximation on TAIEX data
☐	8 compute posteriors, shrinkage factors and out-of-sample errors in Python, passing every divisor explicitly

LEARNING MODULE OVERVIEW

- Bayes' rule for a parameter, (24.1), multiplies a prior density by the likelihood. A beta prior and a binomial record give a beta posterior, (24.2): under a sceptic's Beta(82, 82), eight good months of twelve raise the probability that a manager's monthly chance is at least 0.55 from 10.0% to 15.3%.
- For iid normal returns with a known variance, the posterior mean is the prior mean plus the data's weight c times the gap to the sample mean, (24.3), and precisions add. With a prior standard deviation of 10 points a year, TAIEX's 23.8 years get $c = 0.869$, and the posterior mean is 13.4% a year against the sample's 15.38%.
- A credible interval is a probability statement about the parameter; a confidence interval describes a procedure. Under a flat prior they coincide, 7.8% to 23.0% a year, and $\Pr(\mu \le 0 \mid \text{data}) = 3.8 \times 10^{-5}$ equals LM21's one-sided p-value to the printed digits.
- James–Stein's estimate (24.4) shrinks many means toward their grand mean by an estimated weight. In Experiment 1, with 28 means whose spread equals their noise, it cuts the total squared error from 28.1 to 15.5; the Bayes rule that knows the prior reaches 14.0.
- On the 28 TWSE sector price indices, each sector's first-half mean predicts its second half worse than the grand mean does, 9.8 against 8.5 points a year. The means spread less than 0.945 times their noise, so James–Stein shrinks all the way, even with the common component removed.
- With a normal prior centred at zero, log utility and a known volatility, the Bayes Kelly fraction is c times the plug-in fraction, (24.5). On TAIEX, quarter and half Kelly are the Bayes bets of sceptics whose prior standard deviation is 2.25 or 3.89 points a year.
- The quadratic rule overstates the exact fraction by 0.14 at full Kelly and by 0.10 at $c = 0.869$, while the prior shaves 13%. A prior centred away from zero breaks the fraction reading: centred at 8% a year, the Bayes bet is 0.76 of the plug-in, not $c = 0.50$.

1 INTRODUCTION

LM20 gave a past mean a standard error. Here the unknown mean itself is uncertain: a prior distribution states a belief before the data, and Bayes' rule updates it to a posterior distribution. The posterior mean is a weighted average of the data and the prior, so it shrinks the sample mean toward a target.

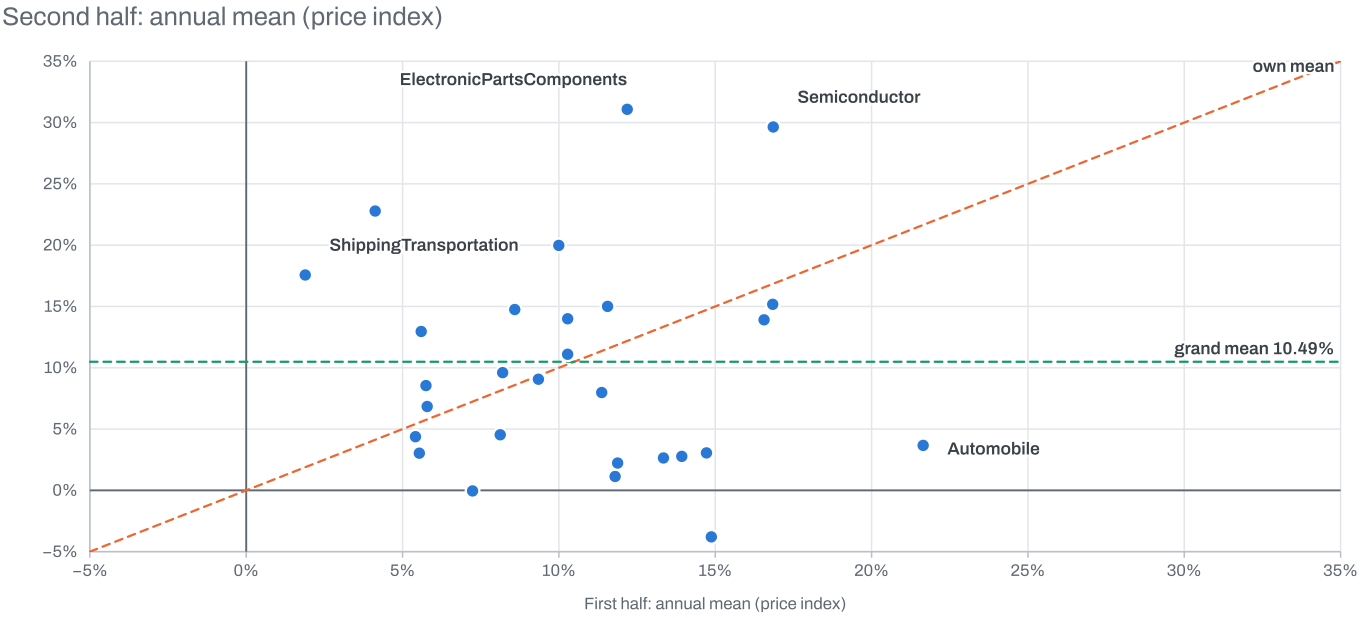
MOTIVATING CASE **Last decade's winners**

Split the 28 TWSE sector price indices of LM10 (U28, no dividends) at LM11's midpoint. From December 2008 to October 2017, Automobile led with a mean return of 21.65% a year. From October 2017 to October 2026 it ranked 20th, at 3.67%. ShippingTransportation went from last, 1.89%, to fifth, 17.57%. Across the 28 sectors, the two halves' means correlate −0.05 (Exhibit 1).

Predict each sector's second half by its own first-half mean, and the typical miss is 9.8 points a year. Predict every sector by the first half's average of all 28, 10.49%, and the miss is 8.5 points.

The TAIEX's total return averaged 15.38% a year over 23.8 years (LM19), and its plug-in Kelly leverage is 4.28 (LM20). LM4 advised betting a quarter to a half of that. Should you believe a past mean, and what would a sceptic have to believe to bet so little?

Exhibit 1: Sector means in two halves



Note: annual mean = A × the mean daily price return, with U28's $A = 245.4794$ for both halves. The halves hold 2,191 returns each: 2 December 2008 → 6 October 2017 and 11 October 2017 → 8 October 2026. Semiconductor was second in both (16.86% → 29.64%), and ElectronicPartsComponents went from ninth (12.19%) to first (31.08%). The typical miss is the root mean square, over the sectors, of prediction minus second-half mean. Price indices (no dividends); out-of-sample. Source: FinMind `TaiwanStockPrice`, sector indices, retrieved 11 October 2026, closes repaired from the 13:30:00 values as in LM10. Computations: `data/lm24_numbers.py`.

Sections 2 to 4 apply Bayes' rule to a parameter and to TAIEX's mean. Sections 5 and 6 shrink many means and test the result on the sectors. Section 7 derives the Bayes Kelly fraction, and Section 8 collects the Python. Shrinking a covariance matrix, the subject of LM30, follows the same idea.

NOTE **Notation for this module**

TR23 is the TAIEX total-return index: $T = 5{,}849$ daily simple returns R_t from 3 January 2003 to 8 October 2026, 23.76 years, $A = 246.12$. U28 has 28 sector price indices and 4,382 returns from 2 December 2008, $A = 245.4794$; in Sections 5 and 6, T is a half's 2,191. Moments divide by T, and s_{u} (divisor $T - 1$) enters only LM21's t-stat. "$A\sigma_0 = 10$ points" is a prior standard deviation (SD) of 10 percentage points a year on the annual mean $A\mu$.

In Sections 3, 4 and 7, μ is TR23's unknown daily mean, and $\sigma = s$ is treated as known. $f^* = \mu/\sigma^2$ is the model's Kelly fraction, $\hat{f} := \bar{R}/s^2$ the plug-in (LM20), $\hat{f}^*$ the exact in-sample root of (4.5) (LM17), f_{B} the Bayes fraction. Local: θ a manager's monthly chance of beating the index; n months, j good ones; x a value of θ (in Section 5's Deep Dive, the argument of the χ^2_ν density); α_0, β_0 a beta prior's parameters; μ_0, σ_0 a prior's mean and SD; c the data's weight; n_0 the prior's worth in observations; densities f_θ, $f_{\theta|\mathrm{data}}$, f_μ, $f_{\mu|\mathrm{data}}$; for $N = 28$ sectors, μ_i, $\bar{R}_i$ the true and sample means, $\bar{\bar{R}}$ their grand mean, $\hat{c}$, $\hat{\mu}_i^{\mathrm{JS}}$ James–Stein's weight and estimate; η LM20's scaled-mean factor; τ a horizon in years; x_k, x_l cells of width Δ in Section 2's sum; a the symmetric beta prior's parameter; $\bar{\varepsilon}$ the noise in $\bar{R}$; ν a chi-square law's degrees of freedom. (16.8)'s slope β is written c.

NOTE **Research hygiene**

TAIEX figures describe the total-return index, 3 January 2003 to 8 October 2026; its posteriors are **in-sample**. Sector figures use U28's price indices (no dividends), estimated on the first half and judged on the second: **out-of-sample**, with LM11's split fixed in advance.

The reverse direction, the lab's thirds and Problem 11 are **stability checks**. The Kelly results and Experiments 1 and 2 are **model** results, not advice, and LM13's manager is **hypothetical**. Every posterior rests on a stated prior, and no test is run.

2 BAYES' RULE FOR A PARAMETER

- ☐ apply Bayes' rule to an unknown parameter, and compute the beta–binomial posterior of a success probability and the data's weight in it

LM13's manager beat the index in 8 of 12 months. LM13 allowed two types: 10% of managers win a month with probability 0.55, the rest with 0.5. But a manager's monthly chance θ can be any number between 0 and 1, so a prior should be a density on $[0, 1]$.

Given θ, months are independent, so by (13.7) a given record of j good months in n has probability $\theta^j(1-\theta)^{n-j}$. As a function of θ, this is the record's **likelihood** $\mathcal{L}(\theta)$ (LM20). Split $[0, 1]$ into short intervals of width Δ. By (13.9) and (13.8), the interval at x_k, with prior probability about $f_\theta(x_k)\Delta$, gets the posterior probability about

$$\frac{\mathcal{L}(x_k)\, f_\theta(x_k)\, \Delta}{\sum_l \mathcal{L}(x_l)\, f_\theta(x_l)\, \Delta}.$$

Dividing by Δ and letting $\Delta \to 0$ turns the sum into an integral; the limit is taken as given here (Wasserman 2004, ch. 11).

THEOREM **Bayes' rule for a parameter**

Let a parameter θ have a prior density f_θ, and let the data have the likelihood $\mathcal{L}(\theta)$, with $0 < \int \mathcal{L}(x) f_\theta(x)\,dx < \infty$. Then the posterior density is

$$f_{\theta \mid \text{data}}(\theta) = \frac{\mathcal{L}(\theta)\, f_\theta(\theta)}{\int \mathcal{L}(x)\, f_\theta(x)\,dx}. \tag{24.1}$$

The denominator does not depend on θ, so constant factors of the likelihood or the prior cancel.

DEFINITION **Beta distribution**

For $\alpha_0, \beta_0 > 0$, the **beta distribution** $\text{Beta}(\alpha_0, \beta_0)$ has the density, for $0 < x < 1$,

$$f_\theta(x) = \frac{\Gamma(\alpha_0 + \beta_0)}{\Gamma(\alpha_0)\,\Gamma(\beta_0)} \times x^{\alpha_0 - 1}(1 - x)^{\beta_0 - 1},$$

with Γ from (15.8). That this density integrates to 1 is stated without proof.

Multiplying the density by x gives a constant times the $\text{Beta}(\alpha_0 + 1, \beta_0)$ density, and the recursion of (15.8) for Γ gives the mean $\alpha_0/(\alpha_0 + \beta_0)$. The same step twice gives the variance

$$\text{Var}\,\theta = \frac{\alpha_0 \beta_0}{(\alpha_0 + \beta_0)^2(\alpha_0 + \beta_0 + 1)}.$$

RESULT **Beta prior, binomial record**

Let $\theta \sim \text{Beta}(\alpha_0, \beta_0)$, and let a record have j successes in n independent trials with probability θ. Then the posterior is $\text{Beta}(\alpha_0 + j, \beta_0 + n - j)$, with mean

$$\begin{aligned} \mathbb{E}[\theta \mid \text{data}] &= \frac{\alpha_0}{\alpha_0 + \beta_0} \\ &\quad + c\left(\frac{j}{n} - \frac{\alpha_0}{\alpha_0 + \beta_0}\right), \\ c &= \frac{n}{n + \alpha_0 + \beta_0}. \end{aligned} \tag{24.2}$$

Proof. By (24.1), the posterior is proportional to $x^j(1-x)^{n-j} \cdot x^{\alpha_0 - 1}(1-x)^{\beta_0 - 1}$, the shape of the stated beta density. Its mean $(\alpha_0 + j)/(\alpha_0 + \beta_0 + n)$ rearranges into (24.2). ■

A prior whose posterior stays in its family is **conjugate**. The **data's weight** c sets n trials against the prior's worth of $n_0 := \alpha_0 + \beta_0$ trials: $c = n/(n + n_0)$.

To match LM13, choose a symmetric prior $\text{Beta}(a, a)$ under which 10% of managers have $\theta \geq 0.55$: $a = 81.96$. The integer 82 still gives 10.0% and keeps every probability an exact fraction. This **sceptic's prior** has standard deviation 0.039 and is worth $n_0 = 164$ months.

EXAMPLE 1

The manager under a sceptic's prior

Under the prior Beta(82, 82), find the posterior after LM13's 8 good months of 12, its mean, the data's weight and $\Pr(\theta \geq 0.55 \mid \text{record})$.

SOLUTION

By (24.2), the posterior is Beta(90, 86), with mean $90/176 = 45/88 = 0.5114$. The data's weight is $c = 12/176 = 3/44 = 0.068$: twelve months weigh 6.8% against the prior's 164. So the record moves the mean from 0.5 by only $\frac{3}{44} \times (\frac{8}{12} - \frac{1}{2}) = \frac{1}{88} = 0.0114$. Exactly, $\Pr(\theta \geq 0.55 \mid \text{record}) = 15.3\%$ (Exhibit 2), against LM13's two-type 13.5%.

EXAMPLE 2

A flat prior, and how long a record convinces the sceptic

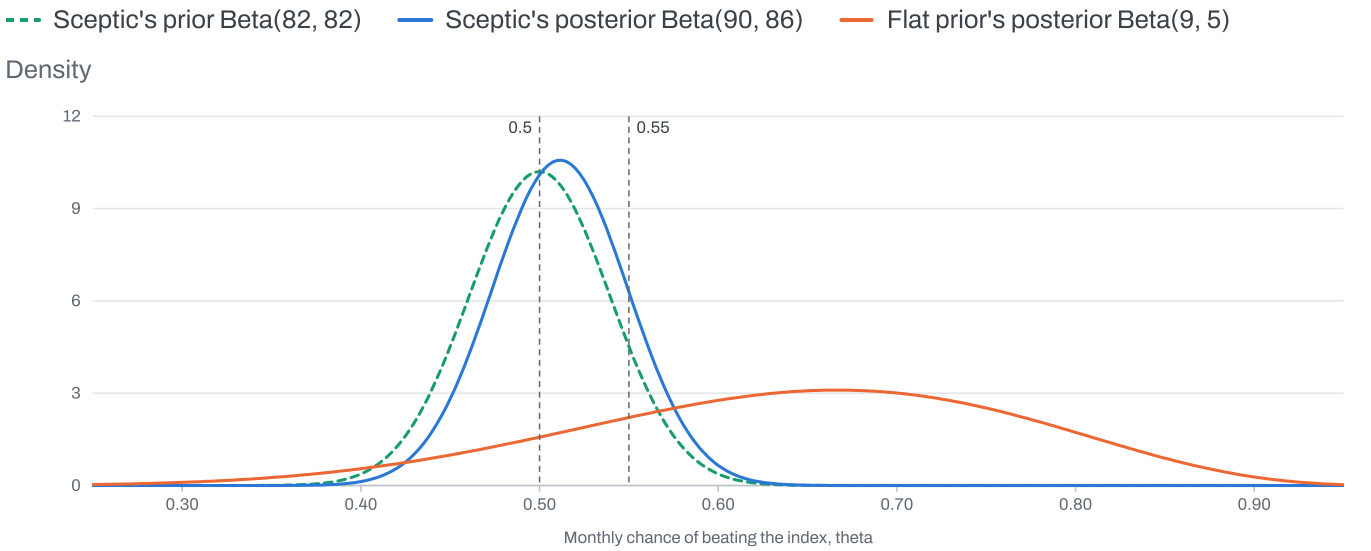
Repeat Example 1 with the flat prior Beta(1, 1). Then find how many months of two good in three raise the sceptic's $\Pr(\theta \geq 0.55)$ above one half.

SOLUTION

The posterior is Beta(9, 5), with mean $9/14 = 0.643$ and $c = 12/14 = 6/7$: the data dominate. Exactly, $\Pr(\theta > 0.5) = 7{,}099/8{,}192 = 86.7\%$ and $\Pr(\theta \geq 0.55) = 77.2\%$. The central 95% of the posterior lies between 0.39 and 0.86.

Under Beta(82, 82), a two-in-three record of n months gives $\text{Beta}(82 + 2n/3,\ 82 + n/3)$. The probability first exceeds one half at $n = 72$ months (51.2%; 49.4% at 69), when $c = 72/236 = 0.305$: six years barely convince the sceptic.

Exhibit 2: A manager's monthly chance θ: prior and posteriors



Note: LM13's hypothetical record, 8 good months of 12. The flat prior's posterior Beta(9, 5) is the likelihood $\theta^8(1-\theta)^4$ scaled to unit area. Exact posterior probabilities of $\theta \geq 0.55$: 15.3% and 77.2% (Example 1, Example 2).

PITFALL **A flat prior is a choice too**

Beta(1, 1) looks neutral, but it puts 45% of managers at $\theta \geq 0.55$, four and a half times LM13's share. Under it, the twelve months dominate the posterior ($c = 6/7$); under the sceptic's prior they hardly move it ($c = 0.068$). State the prior, and say why it was chosen.

導讀 **參數的貝氏推論**

貝氏推論（Bayesian inference）把未知的參數（parameter）當成有分布（distribution）的量：事前分布（prior distribution）描述看資料之前的信念，乘上概度函數（likelihood function）再除以常數，就是事後分布（posterior distribution），即 (24.1)。貝他(β)分布（beta distribution）配上二項資料（binomial data），事後仍是貝他(β)分布，所以它是共軛事前分布（conjugate prior）。事後平均數（posterior mean）是事前平均數與資料頻率的加權平均（weighted average），資料權重（data's weight）$c = n/(n + n_0)$，其中 $n_0 = \alpha_0 + \beta_0$ 是事前分布相當的月數。懷疑性事前分布（sceptical prior）Beta(82, 82) 相當於 164 個月，所以 12 個月的紀錄只占 6.8% 的權重。均勻的事前分布看似中立，其實也是一種選擇。

KNOWLEDGE CHECK 1

A prior Beta(2, 2) meets a record of 3 good months in 4. Find the posterior, its mean and the data's weight.

ANSWER

By (24.2), the posterior is Beta(5, 3), with mean $5/8 = 0.625$. The data's weight is $c = 4/(4+4) = \frac{1}{2}$: the prior is worth four months.

3 THE NORMAL MEAN WITH KNOWN VARIANCE

- [ ] derive the posterior of a normal mean with known variance, and explain why the posterior precision is the prior precision plus the data precision

Now let the parameter be TAIEX's mean daily return μ, with daily returns iid $\mathcal{N}(\mu, \sigma^2)$ and σ known. Expanding each square around $\bar{R}$, the cross term vanishes because $\sum_t (R_t - \bar{R}) = 0$:

$$\sum_{t=1}^{T} (R_t - \mu)^2 = \sum_{t=1}^{T} (R_t - \bar{R})^2 + T(\bar{R} - \mu)^2.$$

So the likelihood $\mathcal{L}(\mu) = \prod_t \varphi\big((R_t - \mu)/\sigma\big)/\sigma$ is a constant times $e^{-T(\bar{R}-\mu)^2/(2\sigma^2)}$. It depends on the data only through $\bar{R}$, and by (24.1) so does the posterior.

Before the data, $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$, and $\bar{R} = \mu + \bar{\varepsilon}$ with noise $\bar{\varepsilon} \sim \mathcal{N}(0, \sigma^2/T)$ independent of μ. So $(\mu, \bar{R})$ is a normal pair by (16.7), with $\operatorname{Var} \bar{R} = \sigma_0^2 + \sigma^2/T$ and $\operatorname{Cov}(\mu, \bar{R}) = \sigma_0^2$ by (16.3) and (16.4). In (16.8), with $X = \bar{R}$ and $Y = \mu$, the slope is $\sigma_0^2/(\sigma_0^2 + \sigma^2/T)$, written c, and the residual variance is $\sigma_0^2(1-c) = c\,\sigma^2/T$.

RESULT **Posterior of a normal mean**

Let the returns $R_1, \dots, R_T$ be iid $\mathcal{N}(\mu, \sigma^2)$ with σ known, and let the prior be $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$. Then

$$\begin{aligned} \mu \mid \text{data} &\sim \mathcal{N}\big(\mu_0 + c\,(\bar{R} - \mu_0),\ c\,\sigma^2/T\big), \\ c = \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2/T} &= \frac{T}{T + n_0}, \qquad n_0 = \frac{\sigma^2}{\sigma_0^2}. \end{aligned} \tag{24.3}$$

Inverting the posterior variance gives $1/\sigma_0^2 + T/\sigma^2$: the **precision** of the posterior, one over its variance, is the prior's precision plus the data's (Gelman et al. 2013, §2.5). The prior is worth n_0 returns.

DEEP DIVE **Completing the square** optional · click to expand

Let m and $v > 0$ be numbers. By (24.1), the log posterior density of μ is a constant plus

$$-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}-\frac{T(\bar{R}-\mu)^2}{2\sigma^2}.$$

This log density is a quadratic in μ with leading coefficient $-\frac{1}{2}(1/\sigma_0^2+T/\sigma^2)$. Its derivative vanishes at

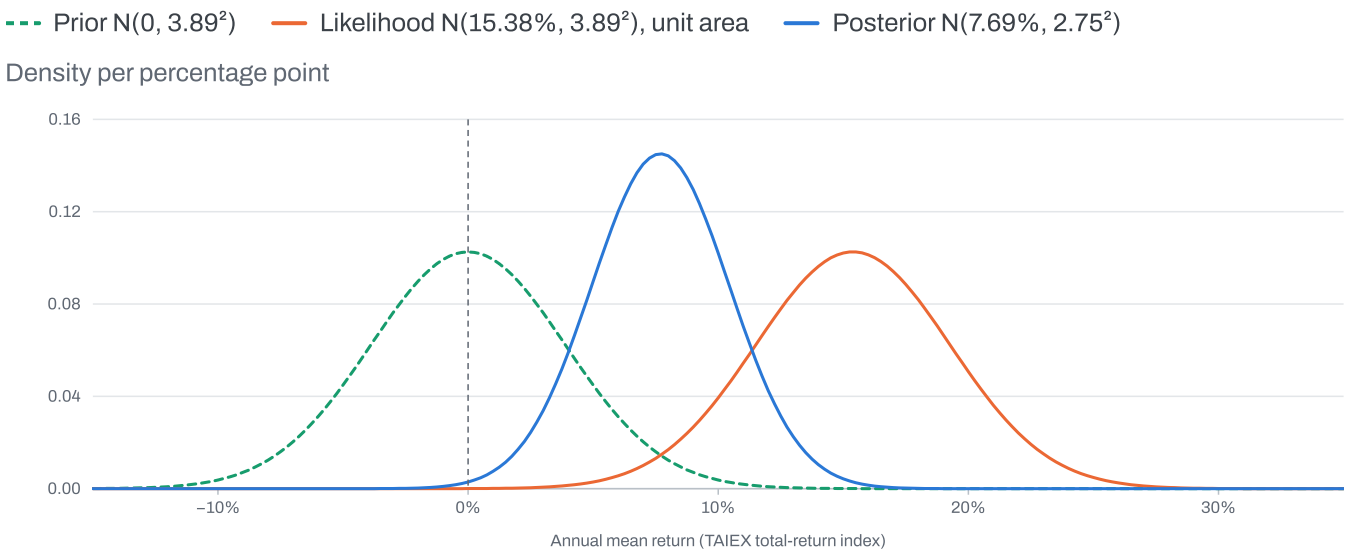
$$\frac{\mu_0/\sigma_0^2+T\bar{R}/\sigma^2}{1/\sigma_0^2+T/\sigma^2}=\mu_0+c\,(\bar{R}-\mu_0).$$

A density whose logarithm is a constant minus $(\mu-m)^2/(2v)$ is the normal density $\varphi\big((\mu-m)/\sqrt{v}\big)/\sqrt{v}$. So the posterior is normal with the mean and variance of (24.3), without (16.8).

On TR23, the model sets $\sigma = s = 1.2084\%$, the normal maximum-likelihood estimate (20.5). LM19's annualization exhibit gives $A\bar{R} = 15.38\%$ and $\sqrt{A}\,s = 18.96\%$ a year, and LM20's (20.2) gives the standard error $As/\sqrt{T} = 3.89$ points. On the annual scale, $c = (A\sigma_0)^2/\big((A\sigma_0)^2+3.89^2\big)$. A prior SD equal to the standard error gives $c = \frac{1}{2}$: the posterior mean halves the sample's, to 7.69%, with SD $3.89/\sqrt{2} = 2.75$ points (Exhibit 3).

For $\mu_0 = 0$, the posterior mean $c\bar{R}$ is LM20's scaled mean $\eta\bar{R}$ with $\eta = c$. By (20.1), $\mathrm{MSE}(\eta\bar{R}) = \eta^2\sigma^2/T + (1-\eta)^2\mu^2$, so the best η needs the unknown μ. Averaged over the prior, μ^2 becomes σ_0^2, and the minimizing η is c. LM20's $\eta^* = 0.940$ is c at a prior SD equal to the mean itself.

Exhibit 3: TAIEX's annual mean: prior, likelihood and posterior at $c = \frac{1}{2}$



Note: TR23, the annual mean $A\mu$. The prior $\mathcal{N}(0, 3.89^2)$, the likelihood $\mathcal{N}(15.38, 3.89^2)$ scaled to unit area and the posterior $\mathcal{N}(7.69, 2.75^2)$, in points a year: the prior is worth exactly the sample, 23.8 years. In-sample. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026. Computations: `data/lm24_numbers.py`.

EXAMPLE 3

TAIEX's mean under a prior of 10 points

Take a prior centred at zero with $A\sigma_0 = 10$ points a year. Find the data's weight, the posterior mean and SD of the annual mean, and the prior's worth in years.

SOLUTION

On the annual scale, $c = 10^2/(10^2 + 3.89^2) = 0.869$. By (24.3), the posterior mean is $0.869 \times 15.38\% = 13.4\%$, and the posterior SD is $\sqrt{0.869} \times 3.89 = 3.6$ points. The prior is worth $n_0 = s^2/\sigma_0^2$ returns, or $n_0/A = (18.96/10)^2 = 3.59$ years of data.

The model assumes normal days, which TAIEX's are not; (17.5) still makes $\bar{R}$ approximately normal. The iid standard error ignores volatility clustering, and robust errors of a mean are LM22's and LM23's.

EXAMPLE 4

Where the prior's variance goes

Check the law of total variance (16.6) for μ and $\bar{R}$ at $A\sigma_0 = 10$ points, on the annual scale.

SOLUTION

By (16.6), $\operatorname{Var}\mu = \operatorname{Var}\big(\mathbb{E}[\mu \mid \bar{R}]\big) + \mathbb{E}\big[\operatorname{Var}(\mu \mid \bar{R})\big]$. Here $\mathbb{E}[\mu \mid \bar{R}] = c\bar{R}$ has variance $c^2(\sigma_0^2 + \sigma^2/T) = c\,\sigma_0^2$. The posterior variance is $(1-c)\sigma_0^2$, whatever $\bar{R}$ is. In points², $100 = 86.9 + 13.1$. Before the data arrive, most of the prior's variance is variance of the coming posterior mean.

PITFALL **A prior SD per day or per year**

The formulas take σ_0 per day, while the text quotes $A\sigma_0$ a year. A prior SD belongs to a mean, so it annualizes by A like the mean, not by $\sqrt{A}$ like a volatility. Dividing 10 points by $\sqrt{A}$ gives a daily σ_0 of 0.637% instead of 0.0406%, and $c = 0.9994$ instead of 0.869: the sceptic disappears. Also, c is a ratio of variances, never of standard deviations (Problem 2).

導讀 **精確度相加**

在變異數（variance）已知的常態模型裡，若平均數（mean）μ 的事前分布（prior distribution）是常態分布（normal distribution），事後分布（posterior distribution）也是常態分布。事後平均數（posterior mean）是 $\bar{R}$ 與事前平均數 μ_0 的加權平均（weighted average），資料權重（data's weight）c 等於事前變異數除以（事前變異數加上 σ^2/T）。事後變異數的倒數稱為精確度（precision），它等於事前的精確度加上資料的精確度 T/σ^2。事前分布相當於 $n_0 = \sigma^2/\sigma_0^2$ 筆報酬，所以 $c = T/(T + n_0)$。加權股價報酬指數 23.8 年的資料，碰上標準差（standard deviation）每年 10 個百分點的事前分布，權重是 0.869，事後平均數由 15.38% 縮為 13.4%。

KNOWLEDGE CHECK 2

With $A\sigma_0 = 10$ points, how many years of TAIEX data is the prior worth? How many years of data would raise c to 0.9?

ANSWER

The prior is worth $(18.96/10)^2 = 3.59$ years. By (24.3), $c = T/(T + n_0) = 0.9$ needs $T = 9n_0$, nine times the unrounded worth, $(18.957/10)^2 = 3.594$: 32.3 years.

4 CREDIBLE AND CONFIDENCE INTERVALS

- [ ] distinguish a credible interval from a confidence interval, and compute both for the TAIEX's mean return

A posterior answers probability questions about μ directly. Below, 1.96 is $\Phi^{-1}(0.975)$, as in LM15.

DEFINITION **Credible interval**

A **95% credible interval** is an interval to which the posterior gives probability 95%. For a normal posterior, the central one is the posterior mean ± 1.96 posterior SD.

LM20's large-sample 95% confidence interval, $\bar{R} \pm 1.96\, s/\sqrt{T}$, says something else. The procedure that builds it covers the fixed, unknown μ in about 95% of samples, while one computed interval either holds μ or misses it. As $\sigma_0 \to \infty$, the prior flattens and $c \to 1$. The credible interval then becomes numerically the confidence interval, and the two differ only in what they claim (Exhibit 4).

Exhibit 4: TAIEX's mean under four priors centred at zero

Prior	$A\sigma_0$ (points)	Worth (years)	c	Posterior mean (% a year)	95% credible interval (% a year)	$\Pr(\mu \le 0 \mid \text{data})$	Bayes fraction $c\hat{f}$
Quarter Kelly	2.25	71.29	0.250	3.8	0.0 to 7.7	0.024	1.07
Half Kelly	3.89	23.76	0.500	7.7	2.3 to 13.1	0.0026	2.14
10 points	10.00	3.59	0.869	13.4	6.3 to 20.5	0.00011	3.72
Flat	∞	0	1	15.4	7.8 to 23.0	0.000038	4.28

Note: TR23, $\sigma = s$, every prior centred at zero; in-sample. The first two rows are defined by $c = \frac{1}{4}$ and $\frac{1}{2}$: their priors are worth three times and once the sample's 23.7645 years. The sceptics and the Bayes fraction belong to Section 7's model. The flat row's interval equals LM20's large-sample 95% confidence interval, and its probability agrees with LM21's one-sided p-value to the printed digits, although LM21 divides by s_{u}. Intervals use $\Phi^{-1}(0.975)$, printed 1.96 as in LM15. Source: as Exhibit 3.

EXAMPLE 5

The chance of a negative mean

Under the flat prior, find the 95% credible interval for TAIEX's annual mean and $\Pr(\mu \le 0 \mid \text{data})$, and compare them with LM20 and LM21. Repeat at $A\sigma_0 = 10$ points.

SOLUTION

With $c = 1$, the posterior is $\mathcal{N}(\bar{R}, s^2/T)$. On the annual scale, $15.38\% \pm 1.96 \times 3.89$ runs from 7.8% to 23.0%, numerically LM20's confidence interval. The posterior probability of a negative mean is

$$\begin{aligned} \Pr(\mu \le 0 \mid \text{data}) &= \Phi\Big(-\frac{\bar{R}}{s/\sqrt{T}}\Big) \\ &= 3.8 \times 10^{-5}. \end{aligned}$$

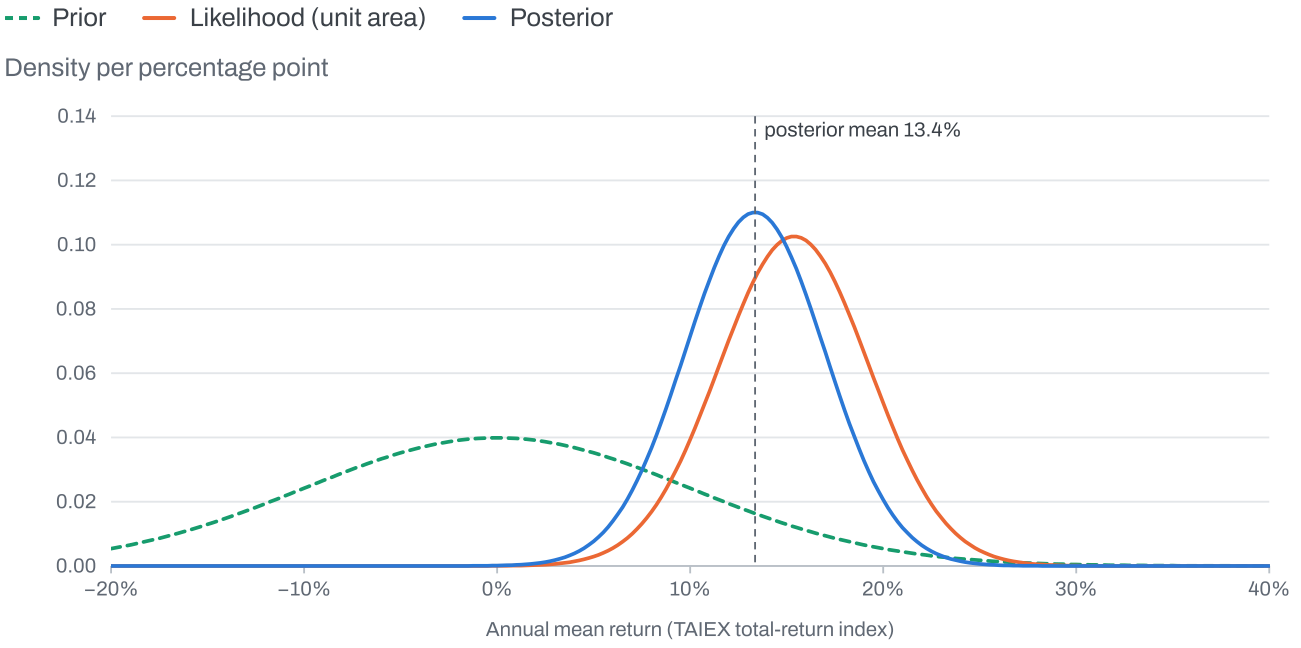
LM21's one-sided p-value, (21.2) at its t-stat 3.96, divides by s_{u} instead of s, and the two agree to the printed digits. Yet the posterior probability is a statement about μ, and the p-value a statement about the data under $H_0 : \mu = 0$.

At $A\sigma_0 = 10$ points, the interval is 6.3% to 20.5% and the probability 1.1×10^{-4} (Exhibit 4). Even under the prior of the first row, the posterior puts 97.6% on $\mu > 0$.

The widget varies the prior SD, the prior mean and the years of data, with TAIEX's annual mean and volatility held at their sample values. It starts at the third row of Exhibit 4.

Prior, data and the Bayes bet

INTERACTIVE



Standard error of the annual mean (points)	Data's weight c	Posterior mean ± SD
3.89	**0.869**	**13.4% ± 3.6**
95% credible interval	**Pr(μ ≤ 0 | data)**	**Bayes fraction · ratio to the plug-in 4.28**
6.3% to 20.5%	**1.1e−4**	**3.72 · 0.869**

Model (24.3) on the annual scale, with TAIEX's annual mean 15.38% and volatility 18.96% (total-return index, 23.76 years) held fixed while the years change: a hypothetical sample. The likelihood is drawn with unit area. The Bayes fraction is (24.5)'s posterior mean divided by A s^2; it equals c times the plug-in 4.28 only when the prior mean is 0. In-sample model results, not advice.

PITFALL **A confidence interval is not a probability for μ**

The sentence "μ lies between 7.8% and 23.0% with probability 95%" belongs to a credible interval, and it needs a prior: here the flat one. Under the prior with $A\sigma_0 = 3.89$ points, the same data give 2.3% to 13.1%. A confidence interval's 95% belongs to the procedure, not to the interval in hand.

導讀 **可靠區間與信賴區間**

可靠區間（credible interval）是事後分布（posterior distribution）的機率（probability）敘述：給定資料與事前分布（prior distribution），μ 落在區間內的機率是 95%。信賴區間（confidence interval）則是程序的性質：重複抽樣（repeated sampling）時，用同一作法建出的區間有 95% 會蓋住固定的 μ，手上這一個區間則只有蓋住或沒蓋住兩種情形。事前分布越平，c 越接近 1，兩者在數字上就越接近，例如加權股價報酬指數年平均報酬（mean return）的 7.8% 到 23.0%。同樣地，$\Pr(\mu \le 0 \mid \text{data})$ 是關於 μ 的機率；p值（p-value）則是在虛無假設（null hypothesis）成立時，關於資料的機率。兩者在這裡數值幾乎相同，意義卻不同。

KNOWLEDGE CHECK 3

(a) Which sentence describes a 95% credible interval? A: "Intervals built this way cover μ in 95% of samples." B: "Given the data and the prior, μ lies in this interval with probability 95%."

(b) Why does the flat prior's $\Pr(\mu \le 0 \mid \text{data}) = 3.8 \times 10^{-5}$ agree with LM21's one-sided p-value to the printed digits, although LM21 divides by s_{u}?

ANSWER

(a) B; A describes a confidence interval.

(b) Since $s_{\text{u}} = s\sqrt{T/(T-1)}$, the factor at $T = 5{,}849$ is 1.0000855. The t-stat changes from 3.956017 with s to 3.955678 with s_{u}, and both give 3.8×10^{-5} (Exhibit 4, last row).

5 SHRINKING MANY MEANS: JAMES–STEIN

- ☐ state the James–Stein estimator with its conditions, and derive its empirical-Bayes reading as an estimated data's weight

Now take N sector means $\bar{R}_1, \dots, \bar{R}_N$ of T returns each, under one prior: the true means μ_i are iid $\mathcal{N}(\mu_0, \sigma_0^2)$. Given the μ_i, the $\bar{R}_i$ are independent, with $\bar{R}_i \sim \mathcal{N}(\mu_i, \sigma^2/T)$ and one common noise variance. By (24.3), with $\bar{R}_i$ as sector i's data, every Bayes estimate $\mu_0 + c(\bar{R}_i - \mu_0)$ moves toward μ_0 by the same weight c.

The data can estimate μ_0 and c themselves: before the μ_i are known, the $\bar{R}_i$ are iid $\mathcal{N}(\mu_0, \sigma_0^2 + \sigma^2/T)$ by (16.7). Their grand mean $\bar{\bar{R}} := \frac{1}{N}\sum_i \bar{R}_i$ estimates μ_0, and the square of their spread estimates $\sigma_0^2 + \sigma^2/T$, the denominator of $1 - c$. Plugging both in gives an **empirical Bayes** estimate (Efron and Morris 1973).

THEOREM **James–Stein estimator toward the grand mean**

Let $N \geq 4$ estimates $\bar{R}_1, \ldots, \bar{R}_N$ satisfy:

1. each $\bar{R}_i \sim \mathcal{N}(\mu_i, \sigma^2/T)$;
2. the $\bar{R}_i$ are independent;
3. the noise variance σ^2/T is common to all of them and known.

Define

$$\begin{aligned} \hat{\mu}_i^{\mathrm{JS}} &= \bar{\bar{R}} + \hat{c}\,(\bar{R}_i - \bar{\bar{R}}), \\ \hat{c} &= \max\Big(0,\; 1 - \frac{(N-3)\,\sigma^2/T}{\sum_i (\bar{R}_i - \bar{\bar{R}})^2}\Big). \end{aligned} \tag{24.4}$$

Then, for every $(\mu_1, \ldots, \mu_N)$, the expected total squared error $\mathbb{E}\sum_i (\hat{\mu}_i^{\mathrm{JS}} - \mu_i)^2$ is below the raw means' $N\sigma^2/T$. Stated without proof (Efron and Hastie 2016, §7.1 and §7.5).

When the means spread several times their noise, $\hat{c}$ is near 1. When their spread (divisor N) is at most $\sqrt{(N-3)/N}$ times the noise, 0.945 for 28 means, $\hat{c} = 0$, and every estimate is the grand mean. James and Stein (1961) shrank toward a fixed point. The max with 0, the **positive part**, stops the weight from turning negative and pushing estimates past the grand mean.

How can a biased rule beat the raw means? In the normal model each $\bar{R}_i$ attains the Cramér–Rao bound (20.7), so no unbiased estimator has a smaller variance. James–Stein accepts bias toward $\bar{\bar{R}}$ and removes more variance than the bias adds, as (20.1) allows. Applied by Efron and Morris (1975) to 18 players' batting averages after their first 45 at-bats of 1970, it had less than half the raw averages' squared error.

DEEP DIVE **Why $N - 3$** optional · click to expand

Under the model, the $\bar{R}_i$ are iid $\mathcal{N}(\mu_0, \sigma_0^2 + \sigma^2/T)$. By Blitzstein and Hwang's Example 10.4.3, $\sum_i (\bar{R}_i - \bar{\bar{R}})^2/(\sigma_0^2 + \sigma^2/T)$ then has the law χ^2_{N-1} of LM16. The χ^2_ν density is $x^{\nu/2-1}e^{-x/2}/\big(2^{\nu/2}\Gamma(\nu/2)\big)$ for $x > 0$ (stated without proof). For $\nu > 2$, the substitution $x = 2y$ in the integral of (15.8) and its recursion for Γ give

$$\begin{aligned} \mathbb{E}\Big[\frac{1}{\chi^2_\nu}\Big] &= \frac{\int_0^\infty x^{\nu/2-2}e^{-x/2}\,dx}{2^{\nu/2}\Gamma(\nu/2)} \\ &= \frac{2^{\nu/2-1}\Gamma(\nu/2-1)}{2^{\nu/2}\Gamma(\nu/2)} = \frac{1}{\nu-2}. \end{aligned}$$

With $\nu = N - 1$, the expectation of $(N-3)(\sigma^2/T)/\sum_i (\bar{R}_i - \bar{\bar{R}})^2$ is $(\sigma^2/T)/(\sigma_0^2 + \sigma^2/T) = 1 - c$. So $1 - \hat{c}$, before the clip at 0, is an unbiased estimate of $1 - c$. The grand mean uses one degree of freedom, and the expectation needs $\nu > 2$: hence $N \geq 4$.

EXAMPLE 6

Experiment 1: shrinkage when the model holds

Simulate 10,000 cross-sections of $N = 28$ true means $\mu_i \sim \mathcal{N}(0, 1)$ and estimates $\bar{R}_i = \mu_i +$ noise, with noise $\mathcal{N}(0, 1)$, so that $\sigma_0^2 = \sigma^2/T = 1$. Use `default_rng([1, 20261011])` : one 10,000×28 array of true means, then one of noise. Compare the average total squared error of the raw means, James–Stein without the clip, its positive part (24.4), and the Bayes rule $\frac{1}{2}\bar{R}_i$, which knows $\mu_0 = 0$ and $c = \frac{1}{2}$.

SOLUTION

The averages are 28.1 for the raw means, 15.5 for James–Stein and for its positive part, and 14.0 for the Bayes rule. Their standard errors are 0.08, 0.04, 0.04 and 0.04, and the mean of $\hat{c}$ is 0.50 (standard error 0.001).

In theory, the raw means' error is $N\sigma^2/T = 28$, and the Bayes rule's is $Nc\,\sigma^2/T = 14$ by (24.3). James–Stein's average, $N - (N-3)/(1+\sigma_0^2) = 28 - 25/2 = 15.5$, is stated without proof (Efron and Hastie 2016, equation 7.14). So estimating μ_0 and c costs 1.5 of the 14 that knowing them saves. (*Model.*)

PITFALL **A target chosen after seeing the data**

The theorem fixes the target in advance: the grand mean, or a number chosen before the data. Shrinking toward the best-looking sector, or toward a value picked because it fits, uses the data twice, and the guarantee is lost. Shrinking regression betas, hierarchical models and Markov chain Monte Carlo are not in this book.

KNOWLEDGE CHECK 4

Four means are 1, 2, 3 and 6, each with noise variance 1. Find $\hat{c}$ by (24.4) and the four James–Stein estimates.

ANSWER

The grand mean is 3, and the squared deviations sum to $4 + 1 + 0 + 9 = 14$. So $\hat{c} = 1 - (4-3) \times 1/14 = 13/14$, and the estimates are $3 + \frac{13}{14}(\bar{R}_i - 3)$: 1.14, 2.07, 3.00 and 5.79 (81/14 exactly).

6 SECTOR MEANS OUT OF SAMPLE

- [ ] evaluate a shrinkage estimator out of sample on two halves of a sample, and explain what correlated estimates do to it

Return to the case, with U28's first half as the data and its second half as the judge, a split that LM11 fixed. Two numbers decide (24.4): the noise of one sector's mean and the spread of the 28 means.

EXAMPLE 7

James–Stein on the sector halves

On U28's first half ($T = 2{,}191$, divisor T), take the common noise variance to be the average of the 28 values s_i^2/T. Find the noise and the spread of the 28 annual means, $\hat{c}$, and the out-of-sample misses at $c = 0$ and $c = 1$. Repeat after removing each day's mean of the 28 returns.

SOLUTION

The noise of one annual mean is A times the root of the average s_i^2/T: 7.18 points. The 28 annual means spread by only 4.50 points (divisor 28). By (24.4), $1 - \hat{c} = 25 \times 7.18^2/(28 \times 4.50^2) = 2.27$, so $\hat{c} = 0$, and every sector is shrunk to the grand mean, 10.49%.

Judged on the second half, the grand mean misses by 8.5 points and the own means by 9.8 (Exhibit 6). The best weight in hindsight, the slope of second-half deviations on first-half ones, is −0.09: the deviations reversed slightly.

Removing each day's mean of the 28 returns leaves the deviations of the 28 means unchanged, but it cuts the noise to 4.79 points. Then $1 - \hat{c} = 1.01$, and $\hat{c} = 0$ again, by a hair. The second half's means spread by 8.49 points, almost twice the first half's: the true means moved, and no first-half estimate could know where.

Example 7 uses one noise variance for all 28 sectors, but the sectors' noise variances differ by a factor of about 4 (half-widths 9 to 19 points in Exhibit 5). Their average, estimated from 2,191 returns, is treated as known. With the total noise, (24.4) gives $\hat{c} = 0$ even if every sector had the smallest noise, 4.8 points (the largest is 9.7), so $\hat{c} = 0$ does not depend on the averaging.

Exhibit 5 is consistent with the zero weight: every first-half interval contains the grand mean.

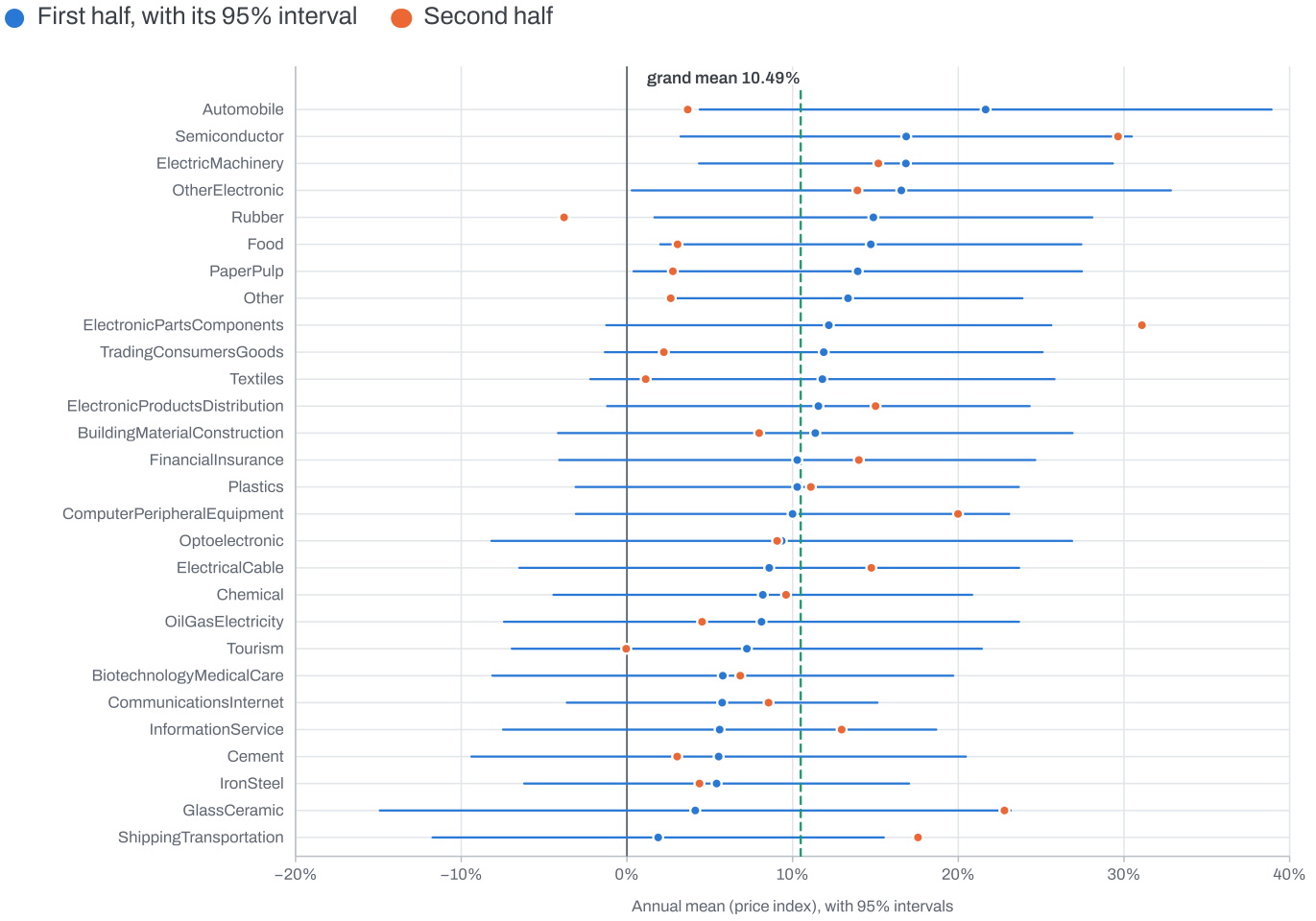
The theorem assumes independent estimates, but sectors share the market's moves. The common part of the noise cancels in the deviations $\bar{R}_i - \bar{\bar{R}}$ yet stays in σ^2/T, so $\hat{c}$ is biased toward full shrinkage. Subtracting each day's cross-sectional mean removes it, and the verdict stands; a market-model alpha (LM23) is another way.

But the 28 subtracted series sum to zero each day, still co-move and have unequal noise, so no dominance theorem covers them. "James–Stein shrank fully" describes these 28 means, not a law. Jorion (1986) carried an empirical-Bayes estimate of means into portfolio choice (LM30).

The misses, the hindsight slope and the case's correlation come from one fixed split of 28 co-moving sectors, so they carry no standard error. The reverse direction and the lab's thirds show how much the

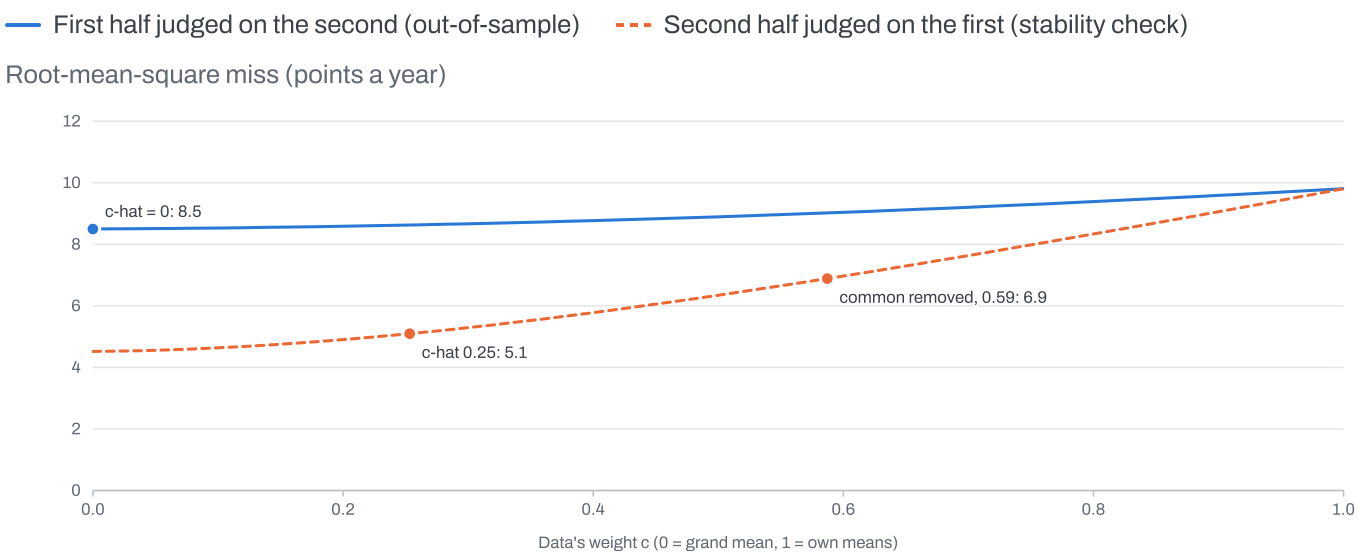
misses move; in the thirds the correlation is 0.13 and 0.24, and the hindsight slope 0.09 and 0.61.

Exhibit 5: First-half sector means with their intervals, and the second half's



Note: the 28 sectors sorted by first-half annual mean, highest at the top. Each first-half mean carries its iid 95% interval $\pm 1.96\, A s_i / \sqrt{2{,}191}$ (LM20), with half-widths from 9 to 19 points. Every interval contains the first half's grand mean, 10.49% (dashed). The intervals are iid and ignore the common component. Price indices (no dividends), U28's halves, $A = 245.4794$; out-of-sample. Source: as Exhibit 1.

Exhibit 6: Out-of-sample miss against the data's weight



Note: the root-mean-square miss over the 28 sectors, in points a year, of $\bar{\bar{R}} + c(\bar{R}_i - \bar{\bar{R}})$ from one half against the other half's means. Forward (first half to second, out-of-sample): 8.5 at $c = 0$ and 9.8 at $c = 1$, with $\hat{c} = 0$ for either noise estimate. Reverse (stability check): 4.5, 5.1, 6.9 and 9.8 at $c = 0$, 0.25, 0.59 and 1, where 0.25 and 0.59 are $\hat{c}$ with the total noise and with the common component removed. Price indices (no dividends), U28's halves. Source: as Exhibit 1.

EXAMPLE 8

The reverse direction

Estimate on the second half and judge on the first. (*Stability check.*)

SOLUTION

The second half's means spread more, so $\hat{c} = 0.25$, or 0.59 with the common component removed. Judged on the first half, full shrinkage still does best: it misses by 4.5 points, against 5.1, 6.9 and 9.8 at $c = 0.25$, 0.59 and 1 (Exhibit 6). In this direction the theorem's weight shrinks too little.

PITFALL **Searching for a split that shows shrinkage**

Other splits, periods or sector groupings would give other weights. Trying them until one shows partial shrinkage turns an out-of-sample test into an in-sample search, and LM21 counts such looks. The split here is LM11's, fixed before any result, and the reverse direction is a stability check, not a second test.

導讀 **收縮到總平均數**

收縮估計量（shrinkage estimator）把每個估計值（estimate）往目標拉近。James-Stein 估計量（James–Stein estimator）的目標是總平均數（grand mean），拉近的程度由資料自己估計，所以它是一種經驗貝氏估計量（empirical Bayes estimator）。關鍵是比較兩個數字：單一產業平均報酬（mean return）的雜訊（noise）有多大，以及 28 個平均報酬彼此差多少。前半段的雜訊每年 7.18 個百分點，彼此的差異只有 4.50 個百分點，所以 $\hat{c} = 0$，全部收縮到總平均數；用後半段檢驗，總平均數的誤差（error）8.5 點，小於各自平均數的 9.8 點。產業報酬彼此相關（correlated），共同成分（common component）在離差（deviation）中抵銷，卻留在雜訊的估計裡；扣掉每天的橫斷面平均（cross-sectional mean）後，結論不變。這是這 28 個平均數的結果，不是定律。

KNOWLEDGE CHECK 5

In how many of the 28 sectors did the mean lie on the same side of its half's grand mean in both halves? How many would independent halves give on average?

ANSWER

13 of 28. Independent halves would put a sector on the same side with probability $\frac{1}{2}$, so 14 on average: no persistence beyond chance.

7 KELLY WITH AN UNCERTAIN MEAN

- ☐ derive the Bayes Kelly fraction under log utility in a stated model, and compute the prior that makes fractional Kelly the Bayes bet
- ☐ explain what breaks the Bayes Kelly result, and measure the size of its quadratic approximation on TAIEX data

LM4 advised betting a quarter to a half of the Kelly fraction. Thorp (2006, §7.3) and MacLean, Thorp and Ziemba (2010) give prudential reasons, and Rising and Wyner (2012) trace the benefit to reduced estimation error. A stated model turns the rule of thumb into a result:

1. the index follows geometric Brownian motion with known volatility σ and unknown drift μ;
2. the prior is $\mu \sim \mathcal{N}(0, \sigma_0^2)$, centred at zero;
3. the investor holds a fixed fraction f and maximizes expected log wealth;
4. borrowing costs nothing, $r_f = 0$.

Under geometric Brownian motion (LM38) a fixed fraction grows at exactly $f\mu - \frac{1}{2}f^2\sigma^2$ a day, and LM42 derives Kelly in continuous time. Given μ, expected log growth over τ years is therefore $(f\mu - \frac{1}{2}f^2\sigma^2)A\tau$. The investor does not know μ and averages over the posterior, in which the growth is linear:

$$\mathbb{E}\Big[\ln\frac{W_\tau}{W_0}\;\Big|\;\text{data}\Big] = \Big(f\,\mathbb{E}[\mu \mid \text{data}] - \tfrac{1}{2}f^2\sigma^2\Big)A\tau.$$

RESULT **Bayes Kelly fraction**

In the model above, the fraction that maximizes posterior-expected log growth, over any horizon τ, is

$$f_{\mathrm{B}} = \frac{\mathbb{E}[\mu \mid \text{data}]}{\sigma^2} = c\,\hat{f}, \qquad \hat{f} = \frac{\bar{R}}{\sigma^2}, \tag{24.5}$$

where the second form uses the prior centred at zero and (24.3). Applying (24.3) to the daily simple returns is an approximation under geometric Brownian motion. For daily returns the same follows from (4.2) without its μ^2 term.

Fractional Kelly is thus the Bayes bet, and its fraction is the data's weight c. Browne and Whitt (1996) derive the general rule under log utility: the bet is linear in the posterior mean.

EXAMPLE 9

TAIEX's Bayes fraction and its exact value

Take $A\sigma_0 = 10$ points. Find f_{B}, the exact fraction on returns shifted to the posterior mean, the posterior SD of the model's $f^* = \mu/\sigma^2$, and the expected log growth.

SOLUTION

By (24.5), $f_{\mathrm{B}} = 0.869 \times 4.28 = 3.72$. The exact Kelly condition (4.5) on TR23's returns, each shifted by one constant to the posterior mean (LM4's band method), gives 3.62. Averaging the sample's growth over the posterior of μ before maximizing gives 3.62 too.

The posterior SD of $f^* = \mu/\sigma^2$ is $\sqrt{c}$ times LM20's mean-only standard error 1.08, or 1.01, so f^* lies between 1.7 and 5.7 with posterior probability 95%.

The quadratic rule overstates the exact fraction by 0.14 at $c = 1$ (4.28 against LM17's $\hat{f}^* = 4.14$), by 0.10 at $c = 0.869$, and by 0.03 at $c = \frac{1}{2}$ (2.14 against 2.11). Keeping the μ^2 term of (4.2), which (24.5) drops, gives $\bar{R}/(s^2 + \bar{R}^2) = 4.27$ at $c = 1$: it explains only 0.01 of the gap. The other 0.13 comes from the days' fat tails and negative skewness (LM19). These shave a few per cent off a fraction; the prior shaves 13% (Exhibit 7).

In the model, the posterior-expected log growth is about 25% a year at f_{B}. The plug-in bet $\hat{f} = f_{\mathrm{B}}/c$ keeps $1 - (1 - 1/0.869)^2 = 97.7\%$ of it, by (4.15) with $1/c$ as its fraction.

EXAMPLE 10

The sceptics behind LM4's rule

Which priors centred at zero make the Bayes bet a quarter and a half of $\hat{f} = 4.28$? Under each sceptic's posterior, compare the expected growth of the Bayes bet and of the plug-in bet.

SOLUTION

By (24.3), $c = \frac{1}{4}$ needs $(A\sigma_0)^2 = 3.89^2/3$, so $A\sigma_0 = 3.89/\sqrt{3} = 2.25$ points: a prior worth three times the sample, 71.3 years. And $c = \frac{1}{2}$ needs $A\sigma_0 = 3.89$ points, a prior worth the sample. After the data, both sceptics put high probability on a positive mean, 97.6% and 99.7% (Exhibit 4), yet they bet little. Size depends on how large the mean is, not only on its sign.

Under the half-Kelly sceptic's posterior, the plug-in bet is twice the Bayes bet. By (4.15) it expects zero growth, against 8.2% a year at 2.14. Under the quarter-Kelly sceptic's posterior, it is four times the Bayes bet and expects −16.5% a year: $1 - (1 - 4)^2 = -8$ times the Bayes bet's 2.06%.

Exhibit 7: TAIEX's Bayes Kelly fraction against the prior's standard deviation



Note: Section 7's model on TR23, prior centred at zero. The quadratic $c\hat{f}$ is (24.5); the exact fraction is the root of (4.5) on TR23's returns shifted to the posterior mean. Dashed: $\hat{f} = 4.28$, LM17's $\hat{f}^* = 4.14$, and $A\sigma_0$ at 2.25 points (quarter Kelly, 1.07), 3.89 (half Kelly, 2.14; exact 2.11) and 10 (3.72; exact 3.62). The gap between the curves comes mostly from the days' fat tails and negative skewness; (4.2)'s μ^2 term adds 0.01 at full Kelly. Model; in-sample. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026.

Four departures break the result:

- **A prior centred away from zero.** The Bayes bet is still $\mathbb{E}[\mu \mid \text{data}]/\sigma^2$, but no longer $c\hat{f}$ (knowledge check).
- **An unknown σ.** Under the conjugate prior of Gelman et al. (2013, §3.3), whose prior SD of μ is proportional to σ, the posterior of μ is a Student t. At $T = 5{,}849$ this matters little. Volatility that changes from day to day is modelled in LM26.
- **Non-normal days.** Fat tails and a negative skewness make the quadratic rule overstate the exact fraction (Example 9).
- **Other utilities.** Power utility, under which fractional Kelly expresses risk aversion, is LM44's subject, and sizing from a bootstrap distribution is LM45's.

> **PITFALL** **The prior is the decision**
>
> Under (24.5) the fraction is c, so choosing the prior is choosing the bet. A prior picked to justify a wanted fraction repeats Section 5's error of a target chosen after the data. State the prior and its worth in years, as Exhibit 4 does, and read the result as a model result, not advice. Haircuts for a backtest's Sharpe ratio are LM33's.

導讀 **分數凱利是懷疑者的貝氏下注**

第 7 節的模型假設波動率（volatility）已知、事前分布（prior distribution）以零為中心、目標是期望對數成長（expected log growth）。在這個模型裡，使事後期望對數成長最大的部位是 $f_{\mathrm{B}} = \mathbb{E}[\mu \mid \text{data}]/\sigma^2 = c\hat{f}$，而且與投資期間長短無關。所以分數凱利（fractional Kelly）就是貝氏下注（Bayes bet），比例恰好是資料權重（data's weight）c。要得到 LM4 的四分之一或一半凱利，懷疑者的事前標準差（prior standard deviation）必須是每年 2.25 或 3.89 個百分點，相當於 71.3 年或 23.8 年的資料。在半凱利懷疑者的事後分布（posterior distribution）下，直接用估計值（estimate）下注就是兩倍凱利，期望成長為零。事前分布若不以零為中心，這個對應就不成立；報酬有厚尾（heavy tail）與負偏態（negative skewness）時，它也只是近似。

KNOWLEDGE CHECK 6

Take a prior centred at 8% a year with $A\sigma_0 = 3.89$ points, so $c = \frac{1}{2}$. Find the posterior mean, the Bayes fraction and its ratio to $\hat{f} = 4.28$. Is the ratio c?

ANSWER

By (24.3), the posterior mean is $8\% + \frac{1}{2}(15.38\% - 8\%) = 11.69\%$ a year. By (24.5), $f_{\mathrm{B}} = 11.69/15.38 \times 4.28 = 3.25$, which is 0.76 of $\hat{f}$, not $c\hat{f} = 2.14$. The fraction reading needs a prior centred at zero.

8 BAYESIAN INFERENCE IN PYTHON

- [ ] compute posteriors, shrinkage factors and out-of-sample errors in Python, passing every divisor explicitly

SciPy's `stats.beta` gives a beta law's mean, quantiles and tail probabilities; for integer parameters, the lab computes them exactly with `Fraction`. The functions below pass every divisor explicitly.

PYTHON

```python
import numpy as np
from scipy import optimize, stats

post = stats.beta(82 + 8, 82 + 4)      # sceptic's prior, 8 good months of 12
print(post.mean(), post.sf(0.55))      # 0.5114, 0.1526


def posterior(Rbar, s, T, sigma0, mu0=0.0):     # (24.3), daily units
    c = sigma0**2 / (sigma0**2 + s**2 / T)
    return c, mu0 + c * (Rbar - mu0), np.sqrt(c) * s / np.sqrt(T)


def james_stein(X):                     # X: T x N daily returns, (24.4)
    T, N = X.shape
    m = X.mean(axis=0)
    noise = (X.var(axis=0, ddof=0) / T).mean()  # common noise variance
    c_hat = max(0.0, 1 - (N - 3) * noise / ((m - m.mean()) ** 2).sum())
    return c_hat, m.mean() + c_hat * (m - m.mean())


def kelly_exact(R, target):             # root of (4.5), returns shifted
    x = R - R.mean() + target
    return optimize.brentq(lambda f: (x / (1 + f * x)).mean(),
                           1e-6, 1 / abs(x.min()) - 1e-6)
```

IN PYTHON **Companion lab**

The Polars-first notebook `LM24_lab.ipynb` rebuilds the module's exhibits and computational examples from TR23's closes and the sector closes, runs Experiments 1 and 2, and solves the computational practice problems. It ends with a stability check on thirds of U28.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

A colleague computes each sector's variance with Polars' `var()` instead of NumPy's `X.var(axis=0)`. What changes in Example 7 with the common component removed?

ANSWER

Polars' `var()` divides by $T - 1$, NumPy's by T unless told otherwise. The noise variance grows by the factor $2{,}191/2{,}190$, so $1 - \hat{c}$ becomes 1.009 instead of 1.008, and $\hat{c}$ stays 0. Pass `ddof` explicitly, so that the divisor is a choice, not a default.

SUMMARY

- Bayes' rule for a parameter (24.1) multiplies a prior density by the likelihood; a beta prior and a binomial record give a beta posterior (24.2).
- Each posterior mean moves from the prior mean toward the data by the data's weight c; for a normal mean with known variance, precisions add (24.3).
- A credible interval is a probability statement about the parameter; a confidence interval describes a procedure.
- James–Stein (24.4) estimates the weight from the spread of many means and beats the raw means under its conditions.
- On U28's halves $\hat{c} = 0$: the 28 sector means carry almost nothing beyond their average.
- In the stated model the Bayes Kelly fraction is $c\hat{f}$ (24.5). Quarter and half Kelly are the bets of sceptics whose priors are worth three times and once the sample.

PRACTICE PROBLEMS

1. A prior Beta(3, 7) meets 4 successes in 6 independent trials. The posterior mean of the success probability is *closest* to:

   A. 0.667

   B. 0.4375

   C. 0.700

2. A prior SD of 6 points a year meets a standard error of the annual mean of 4 points, for a normal mean with known variance. The data's weight c is *closest* to:

   A. 0.60

   B. 0.31

   C. 0.69

3. In Problem 2, the posterior SD of the annual mean, in points, is *closest* to:

   A. 3.33

   B. 7.21

   C. 2.40

4. James–Stein toward the grand mean, (24.4), is applied to $N = 10$ means with common noise variance 1, whose squared deviations from their grand mean sum to 18. The weight $\hat{c}$ is *closest* to:

   A. 0.556

   B. 0.611

   C. 0.389

5. The plug-in Kelly fraction is 3.0, and a prior centred at zero gives the data the weight $c = 0.4$. In the model of (24.5), the Bayes fraction is *closest* to:

   A. 1.20

   B. 1.90

   C. 0.48

6. Under a posterior with $c = \frac{1}{2}$ and a prior centred at zero, the plug-in bet's expected log growth, as a share of the Bayes bet's, is by (4.15):

   A. 75%

   B. 50%

   C. 0%

7. (a) Derive (24.3) by completing the square. (b) Check (16.6) for TAIEX's annual mean at $A\sigma_0 = 5$ points, with LM20's standard error of 3.89 points, to one decimal.

8. Under Section 5's model: (a) derive $\mathbb{E}[1/\chi^2_\nu] = 1/(\nu - 2)$ for $\nu > 2$ from the χ^2_ν density and (15.8). (b) Show that $1 - \hat{c}$ of (24.4), before the clip at 0, is unbiased for $1 - c$. (c) What fails at $N = 3$?

9. Take TR23 with a prior centred at zero and $A\sigma_0 = 5$ points, from the printed 15.38%, 3.89 and 18.96% (annual mean, its standard error, volatility) and $\hat{f} = 4.28$. Find c (three decimals) and the posterior mean, SD and 95% credible interval (one decimal). Then find $\Pr(\mu \le 0 \mid \text{data})$ (one significant figure), the Bayes fraction (two decimals) and the prior's worth in years (one decimal).

10. *(Python)* Experiment 2: with `default_rng([2, 20261011])`, created once, simulate 20,000 cross-sections of 28 means at the prior spreads $\sigma_0 = 0$, 1 and 3, in that order, with noise $\mathcal{N}(0, 1)$. At each spread, draw one 20,000×28 array of true means as σ_0 times standard normal draws (even at $\sigma_0 = 0$), then one of noise. Report the average total squared error of the raw means, unclipped James–Stein and its positive part (one decimal). Compare $N - (N - 3)/(1 + \sigma_0^2)$, and say where the positive part helps.

11. *(Python)* Repeat Example 7 with each day's return of the TAIEX price index subtracted instead of the cross-sectional mean. Report the noise (one decimal), $1-\hat{c}$ (two decimals), $\hat{c}$ and the grand mean's out-of-sample miss (one decimal). Why does that miss rise while the miss of the deviations from each half's grand mean stays 8.5? (*Stability check.*)

12. Repeat LM13's Practice Problem 8, 16 good months in 24, under Beta(82, 82). Find the posterior, its mean (three decimals) and $\Pr(\theta \geq 0.55 \mid \text{record})$ in percent (one decimal), and explain the difference from the two-type 18.02%.

SOLUTIONS

1. **B** is correct. By (24.2), the posterior is Beta(7, 9), with mean $7/16 = 0.4375$; equivalently, $0.3 + \frac{6}{16}(\frac{4}{6} - 0.3)$. A is the data alone, $c = 1$. C, $7/10$, leaves n out of the denominator.

2. **C** is correct: $c = 6^2/(6^2 + 4^2) = 36/52 = 0.69$. A, $6/(6+4) = 0.60$, is a ratio of standard deviations. B, $16/52 = 0.31$, is the prior's weight.

3. **A** is correct. Precisions add, so the posterior SD is $1/\sqrt{1/36 + 1/16} = 3.33$, which is also $\sqrt{c} \times 4$. B adds the variances, $\sqrt{52} = 7.21$. C treats standard deviations as variances, $6 \times 4/10 = 2.40$.

4. **B** is correct: $\hat{c} = 1 - (10-3)/18 = 0.611$. A, $1 - 8/18 = 0.556$, uses $N - 2$, the rule toward a fixed point. C, $7/18 = 0.389$, is $1 - \hat{c}$.

5. **A** is correct: $f_{\mathrm{B}} = 0.4 \times 3.0 = 1.20$ by (24.5). B uses $\sqrt{c}$, 1.90. C uses c^2, 0.48.

6. **C** is correct. The plug-in bet is $f_{\mathrm{B}}/c = 2f_{\mathrm{B}}$, twice the Bayes Kelly bet, and (4.15) with fraction 2 gives $1 - (1-2)^2 = 0$. A, $1 - (1 - \frac{1}{2})^2 = 75\%$, is half Kelly read the wrong way round. B is c, the ratio of the bets.

7. (a) See the Deep Dive of Section 3. (b) Here $c = 25/(25 + 3.89^2) = 0.623$. The posterior mean's variance is $c\,\sigma_0^2 = 15.6$ points^2 and the expected posterior variance $(1-c)\sigma_0^2 = 9.4$; they sum to the prior's 25.0.

8. (a) See the Deep Dive of Section 5. (b) The sum of squared deviations over $\sigma_0^2 + \sigma^2/T$ is χ^2_{N-1}, so the expectation is $(N-3)(\sigma^2/T)\big/\big((N-3)(\sigma_0^2 + \sigma^2/T)\big) = 1 - c$. (c) At $N = 3$, the degrees of freedom are $\nu = 2$, and the integral of $x^{-1}e^{-x/2}$ diverges at 0, so $\mathbb{E}[1/\chi^2_2]$ is infinite. The factor $N - 3$ is 0, so $\hat{c} = 1$: the rule returns the raw means and has nothing to estimate the weight with. Hence $N \geq 4$.

9. On the annual scale, $c = 25/(25 + 3.89^2) = 0.623$. The posterior mean is $0.623 \times 15.38\% = 9.58\%$, or 9.6% to one decimal, and the SD is $\sqrt{0.623} \times 3.89 = 3.07$ points, or 3.1. From these two-decimal values, the 95% credible interval runs from 3.6% to 15.6%, and $\Pr(\mu \leq 0 \mid \text{data}) = 0.0009$. The Bayes fraction is $0.623 \times 4.28 = 2.67$, and the prior is worth $(18.96/5)^2 = 14.4$ years.

10. At spreads 0, 1 and 3, the raw means average 28.0 each time. James–Stein averages 3.0, 15.5 and 25.5, and its positive part 2.1, 15.5 and 25.5, with standard errors from 0.02 to 0.05; theory gives 3, 15.5

and 25.5. The positive part helps when the means hardly spread, because the unclipped weight is then often negative.

11. The noise falls to 5.0 points, $1 - \hat{c} = 1.10$, and $\hat{c} = 0$ again, but the grand mean misses by 12.1 points. Subtracting the index shifts every sector's mean in a half by the same amount, so the deviations from each half's grand mean, and their miss of 8.49, do not change. The sectors' average excess over the index moved from −0.3 to −9.0 points a year, as the index rose 10.8% and then 19.1% a year. No first-half estimate sees that common shift.

12. By (24.2), the posterior is Beta(98, 90), with mean 0.521, and exactly $\Pr(\theta \geq 0.55 \mid \text{record}) = 21.6\%$, against 18.02%. The two-type prior allows only 0.5 and 0.55. The beta prior also puts mass above 0.55, where a record of $16/24 = 0.667$ is more likely, so more posterior mass lies past 0.55.

GLOSSARY

Term	中文	Meaning
Bayesian inference	貝氏推論	Treating an unknown parameter as uncertain, with a prior updated to a posterior by Bayes' rule.
Prior distribution	事前分布	The distribution of a parameter before the data.
Likelihood	概度函數	The probability or density of the data as a function of the parameter.
Posterior distribution	事後分布	The distribution of a parameter given the data, (24.1).
Beta distribution	貝他(β)分布	The law on $(0, 1)$ with density proportional to $x^{\alpha_0 - 1}(1 - x)^{\beta_0 - 1}$.
Conjugate prior	共軛事前分布	A prior whose posterior stays in its family, as beta with binomial data.
Data's weight	資料權重	c: the fraction of the gap from prior mean to data that the posterior mean moves.
Precision	精確度	One over a variance; posterior precision = prior precision + data precision.
Credible interval	可靠區間	An interval to which the posterior gives a stated probability.
Shrinkage estimator	收縮估計量	An estimator that pulls estimates toward a target.
Empirical Bayes estimator	經驗貝氏估計量	A Bayes estimate whose prior is estimated from the data.
James–Stein estimator	James-Stein 估計量	(24.4): shrinks many means toward their grand mean by an estimated weight.
Grand mean	總平均數	The average of several means.
Fractional Kelly	分數凱利	A position cf^* with $c < 1$; in Section 7's model, the Bayes bet $c\hat{f}$.

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